

Data Warehouse Project

Data Warehousing and Visualization Project Report

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Introduction

This report presents the design and implementation of a data warehouse for 2022 US domestic flight departures, built on the public *2022 US Airlines Domestic Departure Data* dataset [1]. The source contains roughly 6.95 million departure events, combining flight operations, carrier information, aircraft characteristics, airport metadata, and hourly weather observations at origin airports.

The analytical goal is to study delays and cancellations. The warehouse connects each departure with time, geography, carrier, aircraft, and weather context, making it possible to locate where disruptions concentrate, identify when they become more likely, and examine which operational or environmental factors help explain them.

The project follows the required data warehouse workflow: source analysis, reconciled database design, data quality assessment, cleaning, dimensional modeling, and ETL implementation. Data quality is treated as a central part of the method, not a preliminary formality. Cleaning choices are therefore justified and logged, particularly when they affect missing weather observations, physical consistency rules, or operational outliers.

LLM support was limited to documentation and review tasks. The main local model was `llama3.2:3b` through Ollama; the notebooks also support optional API-based calls through OpenRouter, but these are treated as review assistance only. Prompts were built from schemas, summary statistics, validation outputs, and JSON-like metadata; raw datasets were not passed to the model. The LLM was used for three specific tasks: interpreting DQA scorecards, interpreting missingness summaries, and translating selected outlier results into operational explanations. No model output was accepted without checking against deterministic notebook results, domain rules, or Phase 1 design artifacts. Where LLM assistance is referenced, the corresponding system and user prompts are shown in the relevant section.

The implementation uses three tools with distinct roles:

- **Python (Jupyter Notebooks)**: data quality assessment, profiling, and ETL implementation.
- **Local and API-based LLMs (Ollama / OpenRouter)**: controlled support for selected interpretation and documentation tasks.
- **PostgreSQL**: hosts both the Reconciled Relational Schema and the final dimensional Data Warehouse.

Domain Analysis

A flight departure is not an isolated event. It can be affected by the departure airport, the time of day, the carrier context, the assigned aircraft, and the local weather at departure. The project therefore treats the departure event as the unit of analysis and enriches it with surrounding operational context. The final dashboards use these dimensions to identify where disruptions concentrate and which contextual signals are associated with higher delay or cancellation pressure, without treating aggregate correlations as direct causal proof.

Three questions drive the analysis. Where and when do disruptions concentrate across the US domestic network? Which carrier, airport, and aircraft profiles show distinct delay or cancellation patterns? And under which weather contexts, especially active weather, low visibility, strong wind gusts, and cloud cover, does disruption risk become more visible?

These translate into three operational objectives:

- Identify the airports, months, weekdays, and departure time bands where delay and cancellation pressure is highest.
- Compare carrier and aircraft profiles as descriptive operational lenses, while avoiding causal claims when the signal is weak or mainly contextual.
- Evaluate which weather contexts are associated with higher disruption risk, without over-interpreting isolated sensor minima.

The analytical output is a set of findings that distinguishes strong evidence from softer contextual patterns. Aggregate statistics often hide these differences, so the dimensional model is designed to support roll-up and drill-down across time, geography, carriers, aircrafts, and weather. In the final Tableau analysis, the strongest patterns are temporal and geographical concentration, followed by clear weather-related signals; aircraft age and manufacturer remain useful for description, but less convincing as standalone explanations.

Dataset Description

The dataset used in this project is the 2022 US Airlines Domestic Departure Data [1], a public Kaggle dataset with approximately 6.95 million departure events. The original files are stored under `data/1-original dataset/`. The source is not already organized as a warehouse: the main table is a large denormalized file, while smaller CSV files provide code descriptions and reference metadata.

`CompleteData.csv` is the primary relation. It contains 40 attributes across

approximately 6.95 million flight records and combines operational data, aircraft metadata, airport geography, and hourly weather observations. The auxiliary files help interpret encoded values and descriptive entities:

- `Carriers.csv`, describing airline carrier codes;
- `Stations.csv`, containing airport and Mesonet station metadata;
- `Cancellation.csv`, mapping cancellation status codes to semantic categories;
- `ActiveWeather.csv`, describing encoded weather conditions.

For selected attributes whose meaning extends beyond the dataset itself, descriptions were checked against aviation and meteorological references. Table 1 lists the main reference groups; it is not a second data dictionary, but explains why certain column descriptions draw on external aviation or weather documentation rather than the Kaggle source alone.

Attribute group	Examples	Reference
Operational flight times	DEP_TIME, TAXI_OUT, AIR_TIME, DEP_DELAY, DISTANCE	BTS airline on-time performance glossary and table documentation [2], [3].
Cancellation categories	CANCELLED, CANCELLATION_REASON	BTS on-time reporting directive, which defines carrier, weather, NAS, and security categories [4].
Aircraft registration	TAIL_NUM	FAA Aircraft Registry and N-number inquiry documentation [5].
Weather observations	WIND_SPD, WIND_GUST, VISIBILITY, TEMPERATURE, ALTIMETER	National Weather Service ASOS material and Iowa Environmental Mesonet ASOS/METAR archive documentation [6], [7].

Table 1: External references used to validate selected source attribute descriptions.

Profiling across the full 6.95 million records established the assumptions underlying the data model. The central one concerns weather observations. Weather attributes were verified to depend uniquely on the combination of station, date, and departure hour:

(MESONET_STATION, FL_DATE, DEP_HOUR)

The dependency was validated across 1,133,884 distinct groups with zero violations, confirming that weather observations are uniquely determined by station, date, and departure hour.

A bijective relationship between `ORIGIN` airport codes and `MESONET_STATION` identifiers was verified across all 375 origin airports: each departure airport maps to exactly one weather station and vice versa.

The same uniqueness property does not hold for destination airports. Profiling identified 311 violations for the dependency:

`DEST → MESONET_STATION`

confirming that weather observations are exclusively tied to the departure airport. In practical terms, the warehouse can safely attach weather context to the origin airport and departure hour, but it should not infer destination weather from the same station field.

Flight cancellations in this dataset are assigned to one of four operational categories, summarized in Table 2.

<code>STATUS</code>	<code>CANCELLATION_REASON</code>
0	Not Cancelled
1	Carrier Cancellation
2	Weather Cancellation
3	National Air System Cancellation
4	Security Cancellation

Table 2: Semantic mapping of flight cancellation status codes to operational categories.

Table 2 is used throughout the project to interpret cancellation status values. In the final warehouse, these categories are represented inside `DIM_JUNK` together with the active weather description.

A dedicated consistency check was run on the 147,830 cancelled flights. Despite their status, all records preserved coherent temporal information. `DEP_TIME` matched `FL_DATE` and `DEP_HOUR` in 100% of cases, and `DEP_DELAY` was never null.

810 of these flights had non-zero `TAXI_OUT`, which suggests that ground procedures had already started before cancellation was registered. No cancelled flight had positive `AIR_TIME`, so the anomaly is limited to pre-departure operations.

Cancelled flights therefore retain useful operational and environmental information and are carried through the cleaning and ETL stages without removal.

Reconciled Database Design

The raw CSV files follow a flat, denormalized export format. The re-engineering step decomposes this structure into relational entities, resolves export-related type issues, and introduces surrogate keys for the transactional tables. At this stage the objective is a stable staging layer that preserves the original granularity, not yet a dimensional structure.

Surrogate keys (`Flight_ID` and `Obs_ID`) were introduced for the two high-volume transactional tables. Reference entities keep their natural keys in the staging layer, which simplifies lookup operations during loading. The `Weather_Observation_ID` foreign key in the `Flight` table is resolved after loading through a join on the natural matching conditions.

The schema organizes the domain into seven tables, as shown in Figure 1. `Flight` and `Weather_Observation` contain the transactional data. `Aircraft`, `Carrier`, and `Station` store descriptive metadata. `Active_Weather` and `Cancellation` are compact lookup tables for domain status codes.



Figure 1: Schema of the reconciled database.

Figure 1 shows the intermediate relational layer used before the warehouse. Its purpose is to separate repeated descriptive entities from the large flight and weather tables while preserving all information needed by the ETL.

Foreign key and unique constraints are intentionally absent from the staging DDL. Enforcing them during bulk ingestion would add validation overhead and dependency-ordering issues. Referential integrity is instead checked in the Python/Pandas pipeline before data is promoted to the warehouse schema.

Several columns that are conceptually boolean or integer-coded are stored as `NUMERIC(3,1)`, including `Low_Level_Cloud` and `Active_Weather_Status`. This comes from Pandas type inference: `NaN` values force an upcast to floating-point during CSV parsing. These inconsistencies are tolerated in staging and corrected during transformation.

Reconciled Database Implementation

The physical schema is implemented in PostgreSQL through `phase 1 - design/2-Reconciled Database.sql`. Since strict constraints are relaxed in staging, the implementation focuses on read performance for the downstream ETL. Indexes are created on the columns most often used in filters and joins: `FL_Date`, `Origin`, `Dest`, `MKT_Unique_Carrier`, and `Cancelled`.

The ingestion sequence follows the logical dependencies without relying on database-level enforcement. Reference tables are loaded first, followed by the high-volume transactional tables. Once `Flight` and `Weather_Observation` are available, a post-load update populates `Weather_Observation_ID` through a join on:

```
Origin = Origin_Airport,  FL_Date = Obs_Date,  Dep_Hour = Obs_Hour
```

At this stage, the schema absorbs the raw data without loss, preserving all artifacts and type inconsistencies that will be addressed in the cleaning phase. The loaded tables serve as input to the data quality assessment pipeline.

Data Quality Assessment

The Data Quality Assessment (DQA) was conducted before transformation. It follows the ISO 25012 dimensions: Completeness, Uniqueness, Validity, Consistency, and Timeliness. Each dimension was evaluated in `phase 2 - etl/1-Data Quality Assessment.ipynb`. The resulting scorecards guided the cleaning decisions. Figure 2 summarizes the pre-cleaning scores.

The DQA figures used in the report are generated from the scorecard CSV files through `phase 4 - report/DQA Figure Generation.ipynb`. In Figures 2 and 3, the color gradient maps lower quality scores to red and higher scores to green; blank cells indicate dimensions that are not applicable to a table.



Figure 2: DQA scorecard: pre-cleaning scores per table and quality dimension.

Figure 2 makes clear that the main issue is not the Flight table itself, which already scores highly, but the Weather_Observation table. Its low uniqueness and consistency scores explain why weather cleaning becomes one of the central steps of the project.

LLM assistance was used to translate the most critical DQA scorecard into an interpretable data-quality risk:

System prompt

You are a Data Quality Auditor for aviation datasets.
Review scorecards and identify critical risks for downstream analytics.

User prompt

Review the DQA scorecard for the Weather_Observations table.
Explain the business risk of low uniqueness and consistency scores.

The response correctly identified the Weather_Observation table as the main DQA risk: duplicated hourly observations and inconsistent meteorological values could bias delay analysis. The remediation strategy, however, was defined entirely by deterministic notebook rules.

Completeness and Missingness

Reference entities (Carrier, Aircraft, Station) exhibited near-perfect completeness. The single exception was a missing `range` value in the Aircraft table (0.02% of records), treated as negligible.

The Weather_Observation table showed a uniform missingness pattern. The relevant meteorological variables had 32,789 missing rows, equal to about 0.47% of the denormalized weather records. Since the count was identical across variables, missingness was row-level rather than column-specific.

A co-occurrence analysis showed that the 32,789 affected flight records mapped to weather rows where all selected weather fields were missing. Only 2,497 of those flights were cancellations, equal to 7.62% of the affected subset and below the 10% safety threshold used for the dropping decision. The mechanism was also investigated statistically. The Jaro-Winkler correlation and a logistic regression for `wind_gust` produced pseudo- $R^2 = 0.031$, while the chi-squared test against `obs_hour` was significant ($p = 0.0$). This suggested MAR (Missing At Random) rather than MCAR. Given the low volume and the limited cancellation exposure within the affected subset, the final decision was to drop these rows after programmatic validation (see Section 7.2).

LLM assistance was used only to frame the interpretation of this missingness pattern. The prompt was intentionally based on aggregate metadata rather than raw rows:

System prompt

```
You are a Data Scientist specializing in aviation databases.  
Interpret statistical missingness in tables like Flights, Weather,  
Stations, or Aircraft. Explain if it is MCAR, MAR, or MNAR based on  
real-world aviation logic. Recommend a DW strategy.
```

User prompt

```
Review the missingness statistics for the Weather_Observations table.  
Provide: classification, operational context, and ETL handling strategy.
```

The model suggested checking the temporal structure before choosing between imputation and deletion. This was useful as a checklist, but the final decision was based on the co-occurrence and temporal-gap checks described above.

Validity-range checks additionally identified residual out-of-range meteorological values. Attributes not used in the warehouse, such as `rel_humidity` and `altimeter`, were allowed to remain nullable in the cleaned reconciled layer. By contrast, residual `VISIBILITY` nulls were imputed before the ETL using the median visibility of the same origin airport, with the global median as fallback. This prevents a nullable analytical measure from being propagated into the final fact table.

Uniqueness and Duplication

The main redundancy was structural. Since the raw data is denormalized at flight-event level, each hourly weather observation is repeated once for every flight departing from the same origin in that hour. The `Weather_Observation` staging table contained 6,954,636 rows instead of 1,133,884 unique hourly readings before cleaning. After removing the weather rows with all selected meteorological fields missing, the cleaned table contained 1,121,382 hourly observations. This required exact deduplication on (`origin_airport`, `obs_date`, `obs_hour`).

A separate fuzzy deduplication analysis was run on the `Station` table using Jaro-Winkler similarity with blocking on `airport_state_code`. It found near-duplicate names such as Chicago O'Hare (ORD) vs. Chicago Midway (MDW) and Phoenix Sky Harbor (PHX) vs. Phoenix-Mesa (AZA). These were confirmed as distinct airports and retained.

Consistency and Validity

Inter-column consistency checks exposed two classes of meteorological anomalies. First, over 800,000 records had sustained wind speed greater than wind gust, which is physically inconsistent. Second, some records had dew point above ambient temperature, which is not valid under standard atmospheric conditions.

Structural validity issues were also recorded. Some conceptually boolean columns were represented as `NUMERIC(3,1)`, as already observed in the staging schema. Another inconsistency involved `cloud_cover`: some rows had value 0 even when cloud-layer flags were active or `lowest_cloud_layer` was below 10,000 ft.

Outlier Assessment

Outlier analysis was used as a decision-support step, not as an automatic deletion rule. IQR, Modified Z-score, and Isolation Forest were applied only to variables that can support the analytical goals: delay, taxi-out, air-time, wind gust, and selected weather context measures. Geographic attributes such as latitude, longitude, and elevation were excluded from the final interpretation because extreme values in those fields mostly describe the real geography of US airports. Route distance was also excluded from the final outlier discussion, since long distances are structural route properties rather than anomalous operational events.

The most important result concerns `dep_delay`. This variable has a long right tail: IQR with $k = 3.0$ flagged 4,067 records, while Modified Z-score flagged 7,661 records. These values are not necessarily errors. Very high delays can represent real operational disruptions, such as severe weather, aircraft rotation problems, or

cascading airport congestion. For this reason, `dep_delay` was kept uncapped in the warehouse.

`taxi_out` and `air_time` were interpreted differently. Their outliers are less useful as extreme analytical events because both measures are constrained by physical and operational plausibility. IQR flagged 2,942 records for `taxi_out` and 2,600 for `air_time`. These variables were therefore selected for controlled winsorization during cleaning, while preserving the flight records themselves.

Weather extremes were treated as contextual information rather than direct data errors. `wind_gust` remains useful for outlier inspection because high gust values can represent disruptive weather. `visibility`, instead, is interpreted mainly through averages or threshold rates rather than as a simple IQR outlier: many readings are at or near zero, and using the minimum alone can overstate the importance of a single sensor value. The multivariate weather check is therefore used to identify unusual weather combinations rather than rows to delete.

The final decision is therefore selective: preserve meaningful operational extremes, cap only measures with physical plausibility limits, and avoid reporting outliers that do not support the project questions.

LLM assistance was used to check whether extreme values should be interpreted as errors or valid operational events:

System prompt

You are an Aviation Operations Analyst.
Translate quantitative outliers into operational realities such as
ATC delays, storms, sensor failures, or data-entry errors.

User prompt

We are analyzing numeric extremes in the Flights table.
Provide likely root causes, data validity interpretation,
and impact on analytics for the selected columns.

The output helped distinguish operational extremes from probable data artifacts. It supported keeping `dep_delay` uncapped, while allowing `taxi_out` and `air_time` to be capped by controlled cleaning rules.

Data Cleaning Strategy

The cleaning logic is implemented through a `CleaningPipeline` class in `phase 2 - etl/5-Cleaning Pipeline.ipynb` and `phase 2 - etl/utils/cleaning.py`. The pipeline is organized into chainable transformation methods. Each method logs row-level and column-level changes through an embedded `AuditLog`. Across the full run, the audit log recorded over 2.3 million transformations, supporting traceability and reproducibility.

Deduplication and Referential Integrity

The `Weather_Observation` staging table was deduplicated on `(origin_airport, obs_date, obs_hour)` after the null-row safety check. The preliminary analysis identified 1,133,884 distinct hourly groups in the reconciled layer; after removing weather observations with all selected meteorological fields missing, the cleaned layer contained 1,121,382 hourly observations. Flight records linked to removed duplicates were then remapped to the surviving canonical observation.

Referential integrity was enforced programmatically. In `phase 2 - etl/utils/cleaning.py`, the method `fix_fk_to_null` sets broken foreign-key values to `NULL`; in `phase 2 - etl/5-Cleaning Pipeline.ipynb`, this method is called for `tail_num` against the cleaned `Aircraft` table. This preserves flight records while marking unresolved aircraft references clearly.

Null Handling

Before dropping weather rows, the pipeline ran a temporal gap check grouped by `origin_airport`. The goal was to verify that removing the weather rows with all selected meteorological fields missing would not create critical breaks in the hourly time series. In this dataset, the operation removed the corresponding 32,789 flight records and reduced the `Flight` table from 6,954,636 to 6,921,847 rows.

Records where weather values exceeded the accepted physical ranges were coerced to `NULL` as part of validity-range enforcement, handled by the general-purpose `fix_validity_range` method. Since `VISIBILITY` is a warehouse measure, its few residual nulls are then imputed with a conservative median-based rule grouped by `origin_airport`.

Outlier Management

Winsorization at the 99th percentile was applied to `taxi_out` and `air_time` to cap physically improbable extreme values while preserving the shape of the operational

distribution.

`dep_delay` was deliberately excluded from winsorization. Multi-day groundings and cascading cancellations are not measurement errors; they are operational events that the warehouse should preserve. Capping them during ETL would remove information that cannot be recovered later.

Business Logic Repairs

Four targeted repairs addressed the structural anomalies identified during the DQA:

Wind gust floor. Where `wind_gust < wind_spd`, the gust value was floored to the sustained speed:

$$\text{wind_gust} = \max(\text{wind_gust}, \text{wind_spd})$$

Swapping the two values was rejected as a strategy, since it would propagate failed sensor readings rather than correct them.

Dew point swap. Where `dew_point > temperature`, the two columns were swapped to restore thermodynamic consistency. This heuristic is appropriate only for small inversions likely caused by sensor transposition; larger deviations would require nullification.

Cloud cover correction. The accepted domain for `cloud_cover` is 0–4, where 0 represents clear sky. Records with `cloud_cover = 0` but active cloud layer flags or `lowest_cloud_layer ≤ 10,000` ft were corrected to `cloud_cover = 1`, the lowest non-clear category. This rule is implemented by `fix_cloud_cover_consistency` in phase 2 - `etl/utls/cleaning.py`.

Cancelled air time. Cancelled flights with positive `air_time` had that field set to NULL by `fix_cancelled_air_time` in phase 2 - `etl/utls/cleaning.py`. This is consistent with the pre-departure nature of cancellations established during the DQA.

Validation Results

The post-cleaning DQA scorecard showed that the main warehouse-relevant quality issues were resolved for both the `Weather_Observation` and `Flight` tables. Figure 3 shows the before/after comparison.

The uniqueness score for `Weather_Observation` moved from 6.29% before deduplication to 100%. The low pre-cleaning score does not contradict the approximately 1.1 million distinct hourly observations: the DQA uniqueness

metric flags row instances duplicated on the business key, while the preliminary analysis counts unique weather groups. Consistency also reached 100% after the wind gust, dew point, and cloud cover repairs. Minor acceptable type warnings, such as `air_time` changing from integer to floating point, are caused by numeric transformations and nullable measure handling; they are non-fatal schema artifacts. Residual nulls in cleaned weather attributes not loaded into the warehouse are documented, while `VISIBILITY` is imputed before fact loading.



Post-cleaning scorecard.

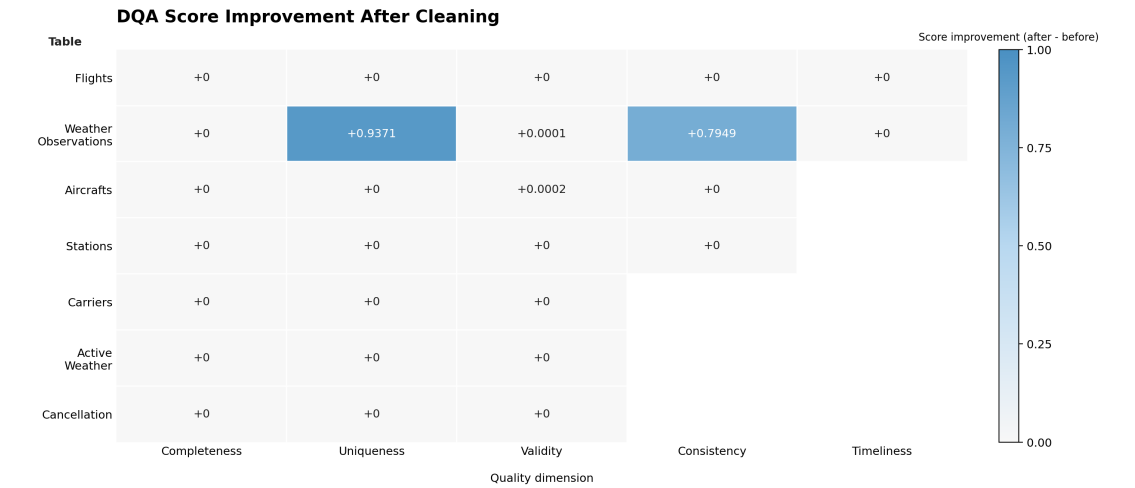


Figure 3: Post-cleaning DQA scorecard and score improvement after the cleaning pipeline.

Figure 3 summarizes the effect of the cleaning phase: the largest gains occur for `Weather_Observation` uniqueness and consistency, while already stable reference tables remain unchanged. The delta view is included to avoid interpreting unchanged high scores as missed cleaning opportunities.

Conceptual Data Warehouse Design

Fact and Grain

The selected fact is the flight departure. The grain is one row per cleaned domestic departure event. This is appropriate because delay, taxi-out, air-time, distance, cancellation status, aircraft, carrier, airport, date, hour, and weather context are all observed at departure level.

Attribute Tree Editing

The initial attribute tree was built from the reconciled `Flights` table, using `FLIGHT_ID` as the root because each record represents one departure event. Carrier descriptions, cancellation reasons, and weather descriptions were grafted from lookup tables. Purely operational identifiers, scheduled timestamps, airport registry codes, and unused weather details were pruned. Aircraft age was derived from `YEAR_OF_MANUFACTURE`. Airport state was retained through the city hierarchy. The edited tree in Figure 4 keeps the attributes useful for OLAP analysis and removes details that do not support the selected questions.

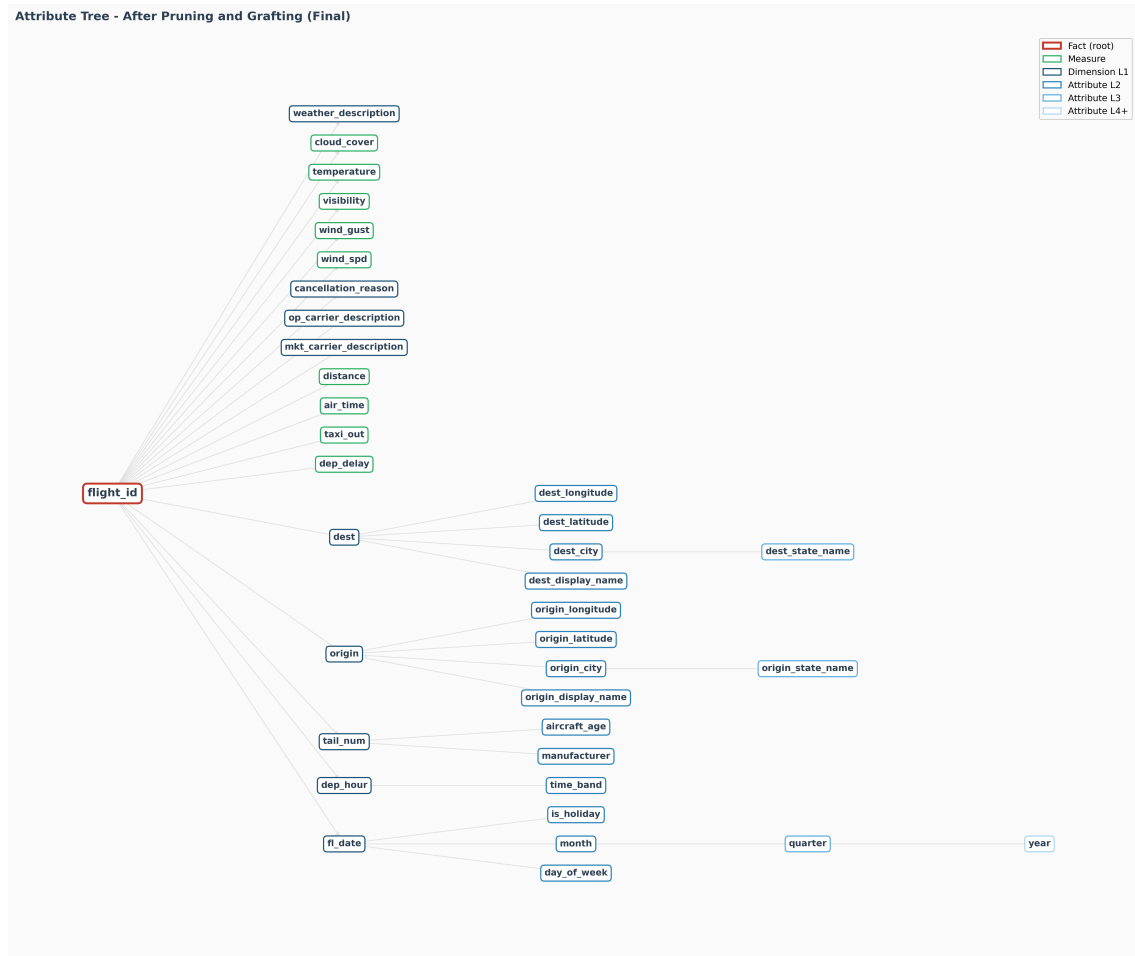


Figure 4: Edited attribute tree after pruning and grafting.

Dimensional Model Design

The Dimensional Fact Model in Figure 5 was derived from the cleaned reconciled schema. Attributes describing the same business perspective were grouped into dimensions: flight date, departure hour, stations, carriers, aircrafts, and a junk dimension for compact categorical indicators. This keeps the fact table focused on measures and foreign keys.

Two dimensions are shared across multiple roles. The station hierarchy is used both for origin and destination analysis, while the carrier dimension is used for marketing and operating carrier perspectives. This avoids duplicating the same descriptive information and keeps roll-up paths consistent. Descriptive attributes are kept close to the dimension they explain: airport name, city, state, latitude, and longitude describe stations; manufacturer and aircraft age describe aircrafts; time band describes departure hour; cancellation reason and weather description are compact descriptors in the junk dimension.

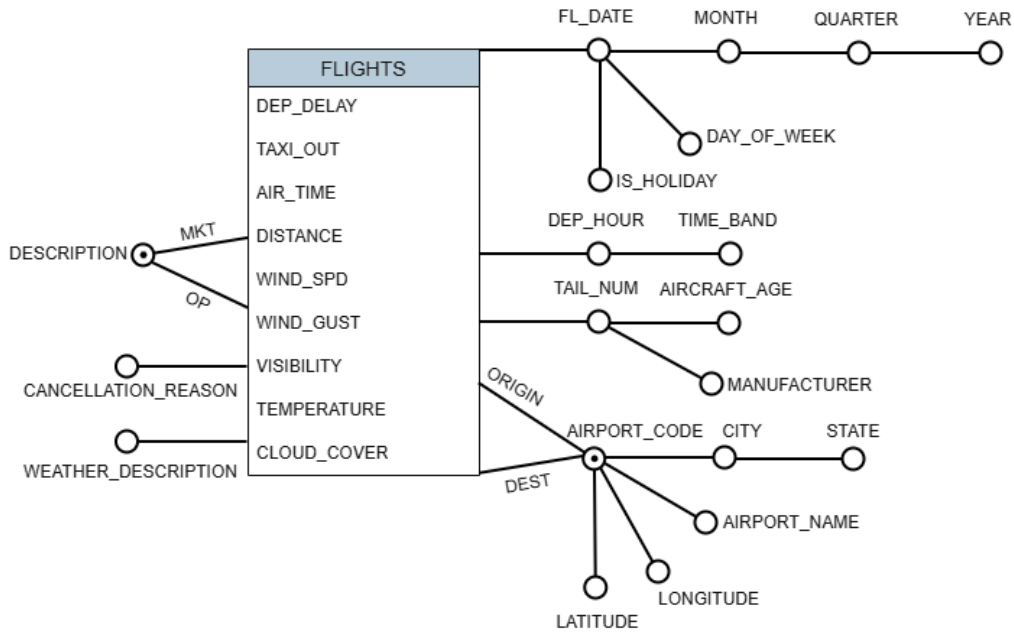


Figure 5: Dimensional Fact Model for the flight departure fact.

Figure 5 shows the selected fact, its measures, and the dimensions available for OLAP analysis. The shared hierarchies make it possible to compare origin and destination geography or marketing and operating carriers without creating separate copies of the same dimension.

Measures

The DFM contains base measures observed at the flight-departure grain: `DEP_DELAY`, `TAXI_OUT`, `AIR_TIME`, `DISTANCE`, `WIND_SPD`, `WIND_GUST`, `VISIBILITY`, `TEMPERATURE`, and `CLOUD_COVER`. Aggregate formulas are not part of the DFM itself; they are documented in the analytical glossary in Table 3. The glossary is defined in **phase 1 - design/5-From DFM to Star Schema.md** and refined in **phase 2 - etl/7-From DFM to Star Schema.ipynb**. Additivity is reported with respect to the underlying base measure semantics: delay minutes are additive, while rates and weather snapshots must be recomputed or aggregated with non-sum operators.

Measure	Formula / Expression	Additivity	Analytical use
# OF FLIGHTS	COUNT(*)	Additive	Volume, map size, denominators.
AVG_DELAY	AVG(DEP_DELAY)	Additive	Delay severity by airport, carrier, route, and weather context.
DELAY_RATE	SUM(CASE WHEN DEP_DELAY > 15 THEN 1 ELSE 0 END) / COUNT(*)	Semi-additive	Main disruption indicator.
CANCELLATION_RATE	SUM(CASE WHEN CANCELLATION_REASON <> 'Not Cancelled' THEN 1 ELSE 0 END) / COUNT(*)	Semi-additive	Cancellation pressure by time band and context.
AVG_TAXI_OUT	AVG(TAXI_OUT)	Semi-additive	Ground operation pressure.
AVG_AIR_TIME	AVG(AIR_TIME)	Semi-additive	Airborne duration context.
AVG_DISTANCE	AVG(DISTANCE)	Semi-additive	Route-distance context.
AVG_WIND_SPD	AVG(WIND_SPD)	Non-additive	Weather context.
AVG_VISIBILITY	AVG(VISIBILITY)	Non-additive	Visibility context without over-emphasizing isolated zero readings.
LOW_VISIBILITY_RATE	SUM(CASE WHEN VISIBILITY > 0 AND VISIBILITY <= 3 THEN 1 ELSE 0 END) / COUNT(*)	Non-additive	Low-visibility scenario analysis.
MIN_TEMPERATURE	MIN(TEMPERATURE)	Non-additive	Cold-weather scenario analysis.
AVG_CLOUD_COVER	AVG(CLOUD_COVER)	Non-additive	Cloud-cover context.

Table 3: Measure glossary derived from the DFM.

Table 3 defines the measures that can be used later in Tableau. It also prevents misleading aggregations: for example, weather readings are not summed, and rates must be recomputed at the selected level of detail.

Logical Data Warehouse Design

The final logical schema follows the design in **phase 1 - design** and is summarized in Figure 6 and Table 4. The fact table is **FLIGHTS**. The dimensions are **DIM_FL_DATES**, **DIM_DEP_HOURS**, **DIM_STATIONS**, **DIM_CARRIERS**, **DIM_AIRCRAFTS**, and **DIM_JUNK**. The DIM prefix is preserved across SQL, notebooks, and generated warehouse CSV files.

FLIGHTS stores one row per departure. It contains surrogate foreign keys to all dimensions and the operational and meteorological measures needed for analysis. The table also includes **LOAD_TIMESTAMP**, a physical ETL metadata column that records when the warehouse batch was generated. **DIM_STATIONS** is reused for origin and destination airport roles, while **DIM_CARRIERS** is reused for marketing and operating carrier roles. This avoids duplicating the same descriptive entities. **DIM_JUNK** groups low-cardinality indicators such as cancellation status, active weather status, and route type.

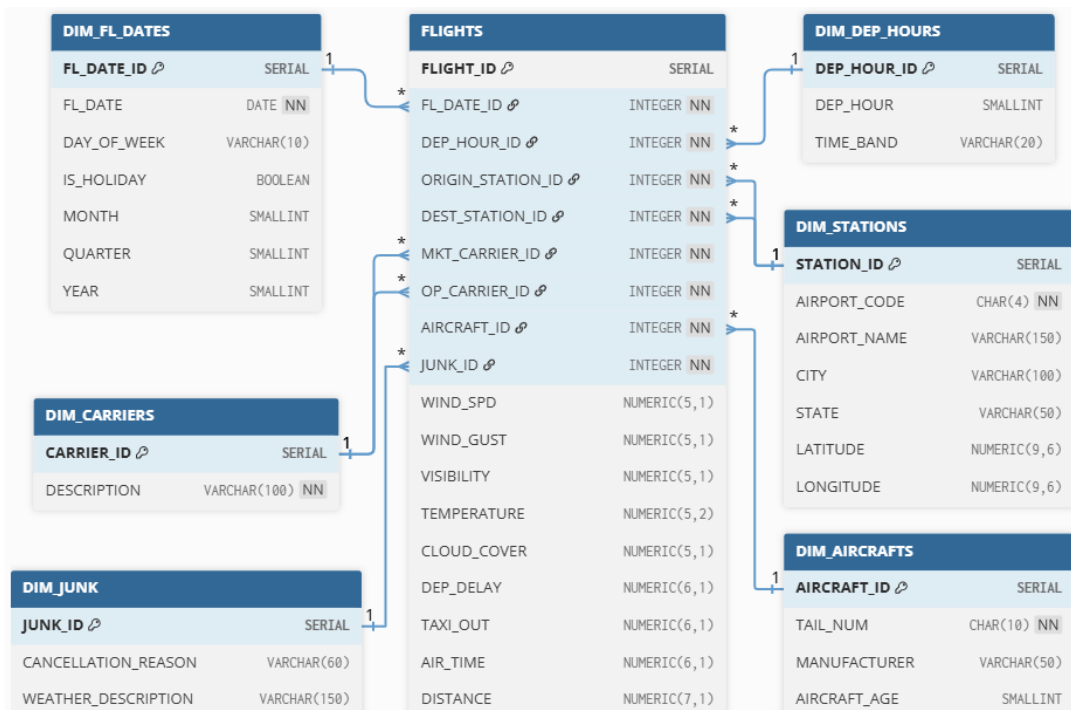


Figure 6: Logical star schema derived from the DFM.

Figure 6 shows the final relational form of the dimensional model. It makes explicit which surrogate keys connect the fact table to each dimension and which dimensions

are reused in more than one analytical role.

Table	Primary key	Rows	Role
DIM_FL_DATES	FL_DATE_ID	365	Calendar dimension.
DIM_DEP_HOURS	DEP_HOUR_ID	24	Departure-hour and time-band dimension.
DIM_STATIONS	STATION_ID	376	Role-playing airport/station dimension.
DIM_CARRIERS	CARRIER_ID	21	Role-playing carrier dimension.
DIM_AIRCRAFTS	AIRCRAFT_ID	6,129	Aircraft registration and age dimension.
DIM_JUNK	JUNK_ID	14	Low-cardinality operational descriptors.
FLIGHTS	FLIGHT_ID	6,921,847	Departure fact table.

Table 4: Final star schema tables and row counts generated by the ETL pipeline.

Table 4 reports the row counts produced by the final ETL run. These counts are used as a consistency check between the generated CSV files in `data/4-data warehouse/`, the ETL metadata in `output/6-ETL/`, and the schema described in this report.

ETL Design and Implementation

The ETL pipeline is implemented in `phase 2 - etl/9-ETL Pipeline.ipynb`. It extracts the cleaned CSV files from `data/3-cleaned database/`, builds surrogate keys for each dimension, resolves role-playing foreign keys, and writes the final dimensional tables to `data/4-data warehouse/`. The implementation is notebook-based and uses reusable Python utilities for configuration, loading, and validation.

Dimensions are loaded first from the cleaned reconciled tables. The fact-loading step then joins cleaned flight records to dimension lookup tables and maps weather measures from the cleaned weather observations. This order allows all foreign keys in `FLIGHTS` to be checked against existing dimensions before the output is considered valid. The generated warehouse contains 6,921,847 rows in `FLIGHTS`. All fact foreign keys are populated, all expected `DIM` tables are present, all fact measures are non-null after the final cleaning pass, and `LOAD_TIMESTAMP` has one batch value per generated table. These checks confirm that the ETL output respects the logical schema defined in Phase 1.

Conclusions

This project produced a complete data warehouse workflow for US domestic flight departures enriched with weather, carrier, aircraft, and station context. The main

result is not only the final set of dimensional CSV files, but the documented path from a raw denormalized source to a cleaned and validated star schema.

The most relevant design decisions were the deduplication and isolation of weather observations, the preservation of meaningful delay extremes, the use of role-playing dimensions for stations and carriers, and the alignment between the Phase 1 design and Phase 2 ETL outputs. The project also demonstrated why data quality assessment must precede dimensional modeling: without the DQA phase, duplicated weather observations, physically inconsistent meteorological values, and misleading outliers would have entered the warehouse silently.

References

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Data Dictionary

This appendix provides the source-level data dictionary used during the preliminary analysis. The descriptions are based primarily on the Kaggle dataset documentation [1], with additional aviation and weather references where necessary. The tables are grouped by original CSV file so that each attribute can be traced back to the raw source. Each table lists the column name, the inferred type observed during profiling, and the meaning used in the design and ETL phases.

CompleteData.csv

Column	Type	Description
FL_DATE	object	Date of flight
DEP_HOUR	int64	Departure hour (0–23)
MKT_UNIQUE_CARRIER	object	Marketing carrier unique code
MKT_CARRIER_FL_NUM	int64	Marketing carrier flight number
OP_UNIQUE_CARRIER	object	Operating carrier unique code
OP_CARRIER_FL_NUM	int64	Operating carrier flight number
TAIL_NUM	object	FAA aircraft registration number
ORIGIN	object	Origin airport code
DEST	object	Destination airport code
DEP_TIME	object	Actual departure timestamp
CRS_DEP_TIME	object	Scheduled departure timestamp
TAXI_OUT	int64	Taxi-out duration in minutes
DEP_DELAY	int64	Departure delay in minutes
AIR_TIME	int64	Flight air time in minutes
DISTANCE	int64	Flight distance in miles
CANCELLED	int64	Cancellation status code
LATITUDE	float64	Airport latitude coordinate
LONGITUDE	float64	Airport longitude coordinate
ELEVATION	int64	Airport elevation in feet

Table 5: Attribute description for the primary source dataset **CompleteData.csv**.

Column	Type	Description
MESONET_ STATION	object	Mesonet weather station identifier
YEAR_OF_ MANUFACTURE	int64	Aircraft manufacturing year
MANUFACTURER	object	Aircraft manufacturer name
ICAO_TYPE	object	ICAO aircraft type designator
RANGE	object	Aircraft range category
WIDTH	object	Aircraft body width category
WIND_DIR	float64	Wind direction in degrees
WIND_SPD	float64	Wind speed in knots
WIND_GUST	float64	Wind gust speed in knots
VISIBILITY	float64	Visibility in miles
TEMPERATURE	float64	Temperature in Celsius
DEW_POINT	float64	Dew point temperature in Celsius
REL_HUMIDITY	float64	Relative humidity percentage
ALTIMETER	float64	Altimeter setting in inches of mercury
LOWEST_ CLOUD_LAYER	float64	Height of the lowest cloud layer in feet
N_CLOUD_ LAYER	float64	Number of cloud layers detected
LOW_LEVEL_ CLOUD	float64	Presence of low-level cloud formations
MID_LEVEL_ CLOUD	float64	Presence of mid-level cloud formations
HIGH_LEVEL_ CLOUD	float64	Presence of high-level cloud formations
CLOUD_COVER	float64	Cloud cover measurement
ACTIVE_ WEATHER	float64	Encoded active weather condition

Table 5: Attribute description for the primary source dataset `CompleteData.csv` (continued).

ActiveWeather.csv

Column	Type	Description
STATUS	int64	Primary key of the weather status
WEATHER_ DESCRIPTION	object	Description of the active weather condition

Table 6: Attribute description for **ActiveWeather.csv**.**Cancellation.csv**

Column	Type	Description
STATUS	int64	Primary key of the cancellation status
CANCELLATION_ REASON	object	Semantic description of the cancellation reason

Table 7: Attribute description for **Cancellation.csv**.**Carriers.csv**

Column	Type	Description
CODE	object	Unique carrier identifier
DESCRIPTION	object	Full airline carrier description

Table 8: Attribute description for **Carriers.csv**.

Stations.csv

Column	Type	Description
AIRPORT_ID	int64	Internal numeric airport identifier
AIRPORT	object	Airport code
DISPLAY_	object	Full airport name
AIRPORT_NAME		
DISPLAY_	object	Full airport city name
AIRPORT_CITY_		
NAME_FULL		
AIRPORT_	object	Full airport state name
STATE_NAME		
AIRPORT_	object	Airport state abbreviation
STATE_CODE		
LATITUDE	float64	Latitude coordinate
LONGITUDE	float64	Longitude coordinate
ELEVATION	int64	Airport elevation in feet
ICAO	object	ICAO airport identifier
IATA	object	IATA airport identifier
FAA	object	FAA airport identifier
MESONET_	object	Associated Mesonet station identifier
STATION		

Table 9: Attribute description for **Stations.csv**.